mod-1 2m

1. What is numpy library for
2. what is pandas library for
3. What does a dataframe object store
4. what is mean imputation in a column in a dataset
5. What is a categorical column in a dataset
6. in a numerical column in a dataset, what is a outlier value
7. What does scaling a numerical column in a dataset mean
8. What does Z-score normalization on a numerical column in a dataset mean

mod-1 5m

1. consider the column : [4,6,-,3,-,10,5,6,-,1] where '–' indicates empty value. Do mean , median, mode imputation on it seperately and write the final mean in each case
2. explain z-score outlier detection. And IQR outlier detection
3. on the column [6,4,7,4,2,6,5,2,3] scale it to fit in range [0,1]
4. do z-score normalization on the column [1,2,3,4,5]
5. create a numerical column of some size between 5 to 8 with integer values, integer mean and integer std deviation. Keep atleast 4 distinct values.

mod-1 12m

1. in a numerical column after doing mean imputation, how does the mean and variance of the column change. Explain with proof
2. (i) say we have a numerical column of values ranging from [a,b], if you want to linearly map them into a different range [x,y] where 'a' maps to 'x' and 'b' maps to 'y'. how to do?

(ii) explain the significance of scaling operation in datapreprocessing

3. say we did a scaling operation where our column values in range [a,b] are scaled to lie between [x,y]. now if we do z-score normalization on both versions. Will we get same result or not? Explain with proof
4. Explain the different stages of data preprocessing. And importance of data preprocessing
5. explain simple user-defined classes and objects in python with examples. Explain attributes and methods, along with the 4 main properties of object oriented programming
6. explain the main differences between numpy array and python lists, how numpy arrays makes life easier
7. explain why/how dataframes make life easy when it comes to handling tabular dataset
8. assume you have a column of 'n' numerical values. Mean is 'M' and std deviation is 'V'. now you want to add a new data point 'x' to this column where x>M and after adding x to the column it is an outlier in the new column. What should be the minimum value of x in terms of M,V,n.

mod-2 2m

1. What is feature engineering
2. correlation coefficient for 2 columns X,Y. What does its value indicate
3. chi2 test to see relationship between 2 categorical columns. What final values you will get, and what does it indicate.
4. how to write a transformation matrix for a linear transformation operation on a point in a nD space
5. what does a eigen vector represent for a given matrix
6. what does a eigen value represent for a particular eigen vector of a matrix
7. for a simple transformation of scaling all points along X direction by factor ‘k’. what are the eigen vectors and eigen values
8. what is the goal of PCA
9. what is dimensionality reduction

mod-2 5m

1. for a simple transformation of rotating all points in XY plane by angle A in clockwise direction. what are the eigen vectors and eigen values. For what values of A we will have any eigen vectors?
2. how to calculate correlation coefficient between 2 numerical columns, what does the value indicate
3. for 2 columns calculate the correlation coefficient X = [5,6,7,8,9,7], Y = [6,10,8,9,7,8]
4. for the matrix calculate the eigen vectors and corresponding eigen values
[[8,3],[2,7]]
5. for the matrix calculate the eigen vectors and corresponding eigen values [[2,2,2],[2,2,2],[2,2,2]]. first try to picture what this transformation means, then try to get the eigen vectors, or just use lambda method

mod-2 12m

1. assume correlation coefficient of 2 columns X,Y is ‘c’ and assume std deviations of X,Y are Sx,Sy and means of X,Y are Mx,My. Now in terms of all these, what will be the correlation coefficient of columns (X,Y+aX). Assume no. of values in each column is ‘n’.
2. (i) assume correlation coefficient of 2 columns (X,Y) is ‘c’. what is correlation coefficient of (X,a*Y+b). explain with proof

(ii) what is correlation coefficient of (X,X)

(iii) if X,Y have a perfect linear relationship, prove that they have +1 or -1 correlation coefficient

3. explain PCA methods to do dimensionality reduction
4. explain the different methods we can do to reduce features, keep important features only

mod-3 2m
1.